

Quick Check

Lesson 7: Plant Hormones: What Controls Growth in Maize and Bean Seedlings?

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.2: Growth and Development in Plants | Lesson 7

A farmer in Kakamega plants bean seeds on the same day after rain. One healthy-looking seed has not germinated because it is still dormant. Which hormone could help break its dormancy and start germination?

- A Absciscic acid
- B Ethylene
- C Gibberellins
- D Auxin from the stem tip only

Which hormone keeps a seed dormant and also causes the stomata of a leaf to close?

- A** Ethylene
- B** Cytokinin
- C** Absciscic acid
- D** Gibberellin

Green bananas bought by a trader in Mombasa turn yellow and soft as they ripen. Which plant hormone controls this ripening?

A Cytokinin

B Gibberellin

C Auxin

D Ethylene

A farmer cuts off the tip of a bean stem and soon side shoots start to grow from the lateral buds. What best explains this?

- A** The tip makes abscisic acid that feeds the side buds, so removing it stops branching
- B** The tip absorbs the rainwater, so removing it leaves more water for the side buds
- C** Cutting the tip makes the plant stop making any hormones
- D** The tip makes auxin that suppresses lateral buds, so removing it lets side buds grow

A student writes: 'Hormones are food for the plant, so a plant with more hormones has more energy to grow.' What is wrong with this idea?

- A** Hormones only work on seeds and never on shoots or roots
- B** Nothing is wrong; hormones are the main food of a plant
- C** Hormones are chemical messengers that control growth processes; they are not food
- D** Hormones are found only in animals, not in plants

Farmers in Kericho root tea cuttings using natural sources such as bulb onion or Aloe extract. Which hormone group do these sources help to supply so that roots form on the cuttings?

- A** Absciscic acid, which keeps the cutting dormant
- B** Ethylene, which ripens the cutting
- C** Gibberellins, which close the stomata
- D** Auxins, which promote the rooting of cuttings

Two maize seeds are planted on the same day with the same rain. One germinates in three days and the other takes longer. What is the best explanation that includes hormones?

- A** Hormones respond to conditions such as water, oxygen and temperature, so seeds can differ in when they germinate
- B** Only seeds without hormones can germinate late
- C** Rain alone decides when every seed germinates, so hormones play no part
- D** Hormones replace water, oxygen and temperature, so these factors do not matter

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).